

Sparse 3° Node Enrichment in Eurasian Monument Corpora

Seth Tenenbaum*
Independent Scholar, USA

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Abstract

We analyze UNESCO Cultural and Mixed World Heritage sites ($N = 1011$) conditional on a specified post-discovery candidate 3° / ~35°E longitude frame. The finding is sparse node-neighborhood enrichment, not whole-corpus alignment: most sites do not fall near harmonic nodes, and the full-corpus Rayleigh periodogram is null ($p = 0.3716$), which rules out global sinusoidal periodicity but not the conditional selected-frame question. Within the candidate frame, the corpus shows modest Tier-A+ enrichment, with stronger dome/stupa behavior: Null A ($p = 0.5429$) locates the dome/stupa signal in the subset's own longitude distribution, while Null C same-footprint backgrounds expect about 11.14–11.84 A+ hits rather than the observed 18. A Normalized Phasor Consensus (NPC) test at the candidate period yields anchor concordance across UNESCO Cultural/Mixed, the nested UNESCO domed-monument subset, and Wikidata Q180987 stupas ($K = 0.9857$, perm- $p = 0.0119$), with a consensus grid line within 0.018° of 35°E; OWTRAD Silk Road route nodes do not share this anchor. The result is best read as sparse, structured node-neighborhood enrichment within a specified 3° / ~35°E monument-corpus frame, supported by conditional diagnostics and monument-corpus anchor concordance, while bounded by the null global periodogram, marginal anchor-discovery correction, geographic dependence, OWTRAD non-portability, and the absence of evidence for universal periodicity, historical intentionality, metrology, or a unique Gerizim origin.

Keywords: template alignment, spatial statistics, Levant, UNESCO World Heritage, 3-degree harmonics, selected-frame analysis, circular statistics, monument distribution, anchor consensus

1 Introduction

This study asks a conditional spatial-statistical question. Given a specified post-discovery candidate 3° / ~35°E longitude frame, do externally curated monument corpora show sparse overrepresentation near harmonic-node neighborhoods, and is that overrepresentation structured rather than merely a consequence of whole-corpus longitude unevenness? The discovery history of the frame is not the evidentiary

*ORCID: [0009-0008-5797-2498](https://orcid.org/0009-0008-5797-2498). Correspondence: seth@fourthtemple.com

object. The object is the behavior of UNESCO Cultural and Mixed World Heritage sites ($N = 1011$) and related monument datasets once the candidate frame is stated.

The candidate frame is not presented as pre-registered or parameter-free. It is evaluated here as a stated post-discovery candidate: the evidentiary claim is conditional on that specification and rests on whether multiple diagnostics and related monument corpora exhibit coherent structure under the same frame.

The conditional audit yields a limited but statistically nontrivial pattern. Within the candidate frame, 132 of 1011 UNESCO sites fall in Tier-A+ (13.1% versus the 10% geometric null). This is a sparse-node enrichment result: most sites do not align, and the full corpus is not globally periodic. The full-corpus Rayleigh periodogram on raw longitudes is null ($p_{\text{perm}} = 0.3716$), so the paper does not claim smooth sinusoidal periodicity, whole-corpus alignment, a universal grid, historical intentionality, or a unique Gerizim origin.

The subset and cross-corpus results ask whether the sparse overrepresentation has structure within Eurasian monument geography. Domed/stupa-related monuments show stronger node-neighborhood behavior. Null A ($p = 0.5429$) locates this enrichment in the dome/stupa sites' own longitude distribution; that is, the signal is carried by the monument subset's geography rather than imposed uniformly across the UNESCO corpus. Null C asks whether the broader UNESCO background occupying the same longitude footprint produces comparable A+ counts; across the tested windows it expects about 11.14–11.84 A+ hits rather than the observed 18, with marginal same-footprint evidence for residual subset structure.

The Normalized Phasor Consensus (NPC) test adds an anchor-concordance view at the candidate 3° period. UNESCO Cultural/Mixed sites, the nested UNESCO domed-monument subset, and the externally curated Wikidata Q180987 stupa corpus yield similar recovered anchors ($K = 0.9857$, $\text{perm-}p = 0.0119$), with a consensus grid line 0.018° from 35°E . These datasets are historically and geographically related rather than geography-independent, but that is appropriate to the claim: the target phenomenon, if real, is a structure within Eurasian monument geography itself. OWTRAD Silk Road route nodes provide a negative control at the same period and do not share the monument NPC anchor.

1.1 Evidentiary Hierarchy

The analyses are organized around five inferential roles so that primary claims, boundary conditions, and exploratory material are not assigned the same weight.

A. Primary conditional frame evidence. Under the selected $3^\circ / \sim 35^\circ\text{E}$ harmonic-deviation representation, the UNESCO corpus shows modest full-corpus Tier-A+ enrichment, the dome/stupa subset shows stronger A++ and A+ behavior, and harmonic-band / cluster-asymmetry tests characterize the distribution of sites around A+ harmonics. These are conditional selected-frame results, not claims of universal alignment.

B. Anchor-concordance evidence. NPC asks whether multiple monument datasets yield similar 3° anchors when each corpus contributes a normalized phasor direction. The three-dataset monument result supports monument-corpus anchor concordance, while recognizing that the UNESCO dome subset is nested within UNESCO and that all three datasets are geographically related.

C. Geographic conditioning. Null A locates the dome/stupa signal in the subset's own longitude geography. Null C asks whether the broader UNESCO background occupying the same longitude footprint produces comparable A+ counts. Together they treat geography as part of the object being studied, while still testing

whether same-footprint background expectations are sufficient.

D. Boundary conditions. The Rayleigh periodogram tests for global sinusoidal periodicity in raw longitudes and is null; the global anchor-discovery correction is marginal; and OWTRAD does not share the monument NPC anchor. These results prevent an invariant-period, whole-corpus, or generic Eurasian-spatial-dataset interpretation.

E. Exploratory and descriptive material. Religion subgroup analyses, the temporal gradient, mound-evolution extensions, the founding classifier, and other sensitivity results are reported to characterize the pattern and generate hypotheses. They are not load-bearing evidence for the primary conditional claim.

Prior longitude-alignment studies were criticised for investigator-assembled site lists (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). This study uses externally curated corpora and fully deposited code (Tenenbaum, 2026). Its evidentiary standard is transparent conditional testing under a clearly specified candidate frame, not a claim that the pattern was discovered in a parameter-free or geography-independent way.

2 Background

2.1 Prior Work on Spatial Periodicity and Sacred Landscapes

Statistical evaluation of spatial periodicity in monument distributions has a long and contested history. Thom’s “megalithic yard” (Thom, 1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique (Freeman, 1976). Magli (Magli, 2013, 2016) has applied Monte Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties.

2.2 Reference Phase and Candidate Anchors

The harmonic deviations in this study are computed relative to a 3° phase in the Levantine ~35°E corridor. Mount Gerizim (35.269°E) is used as the named reference coordinate for that phase because of its independently documented *tabbur ha-aretz* (“navel of the earth”) role in Samaritan tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989). That background motivates a candidate point within the corridor; it is not treated as statistical evidence that Gerizim is the unique origin of a longitude frame.

Caveat on anchor identification. The statistical analysis cannot uniquely identify which physical location, if any, historically defined a phase origin. Any longitude in the x.25°E band produces the same selected-frame enrichment, and any UNESCO site within the A+ tier ($\lesssim 16.6$ km) of the Gerizim-labeled co-grid could be used to reconstruct an essentially identical representation. The primary object is therefore the 3° phase and ~35°E corridor, with Gerizim functioning as one historically motivated label for that phase.

A site-as-anchor ranking audit (Supplementary Audit S6) is consistent with this interpretation: the top-ranked competing anchors, including the Silk Roads: Zarafshan-Karakum Corridor (62.21°E, deviation 0.061° from Gerizim harmonic

$n = 9$), the Birthplace of Jesus (35.21°E), and Old City of Sana’a (44.21°E), all lie within the A+ tier of the Gerizim-labeled co-grid and are therefore co-grid points rather than independent alternatives.

A global anchor sweep (§5.3) likewise shows that the result is not sensitive to the exact coordinate: every longitude in the x.25°E band produces the same enrichment, and both Gerizim and Jerusalem (35.232°E, ≈ 4.1 km south) place the corridor in the global top 2.3% of 36,000 trial anchors. Mecca ($p \approx 0.2825$, ns), Mt. Meru ($p \approx 0.3956$, ns), and Mt. Kailash ($p \approx 0.3956$, ns) yield no significant enrichment under the same procedure.

Gerizim’s coordinate is from the UNESCO tentative list (ref. 5706, DMS E35 16 08, State of Palestine, 2012; <https://whc.unesco.org/en/tentativelists/5706/>) and is not inscribed. Jerusalem is an inscribed corpus member, self-excluded when used as anchor.

3 Data and Methods

3.1 Corpus Construction

The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre, 2024) contains 1248 inscribed nominations. UNESCO designates each site *Cultural* (criteria i–vi), *Natural* (criteria vii–x), or *Mixed*; sites with cultural heritage alongside natural values receive Mixed and are included here. Retaining Cultural and Mixed (–235 Natural-only) and excluding 2 entries lacking geocoordinates (both linear route networks) yields $N = 1011$ sites.

3.2 Geodetic Framework

The selected representation uses a 3° phase in the Levantine $\sim 35^\circ\text{E}$ corridor, operationalized by the Mount Gerizim coordinate at 35.269°E (see §2.2). Gerizim is not inscribed and does not appear in the standard corpus. Jerusalem is inscribed and scored as a regular site in Tests 1–4 (A+ at 4.1 km under the Gerizim-labeled phase). For sensitivity sweeps (§5.3), Gerizim is added as a synthetic entry so each anchor self-excludes symmetrically.

For a site at longitude λ , the deviation from the nearest 3° harmonic is $\delta = |\lambda - 35.269| \bmod 3$ (folded to $[0, 1.5]$). The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim 16.6$ km for Tier-A+) are equatorial approximations computed as $\delta \times 111$ km/°, becoming smaller at higher latitudes; the angular criterion is consequently conservative at high latitudes. No latitude correction is applied.

The geometric null for Tier-A+ is $2 \times 0.15/1.5 = 10\%$, assuming uniform longitudes. This serves as a convenient reference, but the primary tests are simulation-based null models (§5.5) that preserve the empirical longitude distribution. A Gaussian KDE produces 101.1 ± 9.6 expected A+ ($\sim 10\%$), indicating the geometric null is not materially inflated by longitude clumping.

3.3 Tier Classification

Tier boundaries are derived from the cross-corpus von Mises dispersion scale $\hat{\sigma} = 0.1922$ (stupa corpus, $\hat{\kappa} = 6.17$; see “Von Mises cross-corpus characterisation” below). The three A-tiers correspond to $0.25\hat{\sigma}$, $0.50\hat{\sigma}$, and $1.00\hat{\sigma}$ respectively.

Tier A++ $0.25\hat{\sigma}$; $\delta \leq 0.06$ ($\lesssim 6.66$ km); $P_0 = 4\%$.

Tier A+ $0.50\hat{\sigma}$ (primary threshold); $\delta \leq 0.15$ ($\lesssim 16.6$ km); $P_0 = 10\%$.

Tier A $1.00\hat{\sigma}$; $\delta \leq 0.3$ ($\lesssim 33$ km); $P_0 = 20\%$.

Tier B Middle zone, not near any harmonic or inter-harmonic midpoint.

Tier C/C−/C−− Symmetric mirror of A/A+/A++ measured from the nearest inter-harmonic midpoint.

Von Mises cross-corpus characterisation. We fitted $f(\theta \mid \kappa) \propto e^{\kappa \cos \theta}$ (von Mises + uniform mixture) to the harmonic phase deviations of two external corpora: OWTRAD Silk Road nodes ($N = 1674$; $\hat{\kappa} = 9.58$) and Wikidata Q180987 stupas ($N = 229$; $\hat{\kappa} = 6.17$). The stupa corpus yields $\hat{\sigma} = \kappa^{-1/2} = 0.1922$; $0.25\hat{\sigma} = 0.048$ (A++), $0.50\hat{\sigma} = 0.096$ (A+), and $1.00\hat{\sigma} = 0.1923$ (A) match the tier thresholds to within rounding (Table 1). The N -weighted pooled $\hat{\sigma}$ is 0.1588. Both external corpora are concentrated in the Eurasian corridor, so their use here fixes a cross-corpus angular scale within the same geographic domain rather than providing geography-independent validation.

Table 1: Tier boundaries as multiples of the cross-corpus von Mises dispersion $\hat{\sigma} = 0.1922$ ($\hat{\kappa} = 6.17$, Wikidata Q180987 stupa corpus, $N = 229$).

$\hat{\sigma}$ multiple	Threshold	P_0
$0.25\hat{\sigma} = 0.048$	0.06 (≈ 6.66 km)	4%
$0.50\hat{\sigma} = 0.096$	0.15 (≈ 16.6 km)	10%
$1.00\hat{\sigma} = 0.1923$	0.3 (≈ 33 km)	20%

3.4 Test Populations

Keyword populations are approximate. The raw sweep tolerates known false positives while the context-validated variant (Test 2x) applies sentence-level gating for greater precision. Both populations are reported.

Test 1 (full corpus, primary). Full corpus ($N = 1011$), no keyword filter; binomial test of whether the global A+ rate exceeds the 10% geometric null.

Test 2 (dome/spherical, primary). 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site’s name, XML description, and extended OUV text ($N = 90$). This raw sweep is over-inclusive: it admits sites where the keyword refers to geological or vernacular features rather than monumental architecture. The context-validated population ($N = 83$) is reported alongside all primary results; it yields stronger significance, consistent with the rejected sites being non-architectural false positives.

Test 2b (mound evolution, sensitivity variant). Extends Test 2 by adding earthen mound-stage keywords (*tumulus*, *tumuli*, *barrow*, *barrows*, *kofun*). Not counted separately in the Bonferroni family.

Test 3 (cluster asymmetry, primary). Full corpus ($N = 1011$), no keyword filter; tests whether A+ harmonics host more total sites than non-A+ harmonics, independent of any morphological keyword.

Test 4 (temporal gradient, descriptive). Cochran-Armitage trend test on the full corpus based on year of inscription into the UNESCO corpus, no keyword filter.

Test 5 (founding classifier, post-hoc). A post-hoc keyword classifier identifies site types beyond domed monuments that fall on harmonic nodes; it carries no weight in the primary evidential chain (Appendix A.2).

3.5 Multiple-Comparisons Protocol

Full reported family ($k = 44$). The multiple-comparisons headline treats the 44 tests in the FDR audit as the correction family. Six results survive Bonferroni at $k = 44$: dome/spherical A++ (binomial $p_{\text{adj}} = 0.0416$, with the Null A geographic-conditioning result discussed in §4.3), full-corpus A+ ($p_{\text{adj}} = 0.0465$), cluster permutation ($p_{\text{adj}} = < 0.0001$), harmonic Mann-Whitney ($p_{\text{adj}} = 0.0012$), sequential-peak permutation ($p_{\text{adj}} = 0.0279$), and evolution combined A++ (binomial $p_{\text{adj}} = 0.0208$).

Non-primary Tests 1–3 sensitivity set.

- **Test 1** (full corpus, $N = 1011$): global enrichment above 10% null.
- **Test 2** (domed/spherical monuments, $N = 90$): morphological sub-population enriched above 10% null.
- **Test 3** (cluster asymmetry, full corpus, no keyword filter): A+-bearing harmonics host more total sites.

This three-test grouping was not pre-specified and is reported only as a raw-p-value sensitivity summary. The primary multiple-comparisons report is the full $k = 44$ audit above.

Other tests. Test 4 (temporal gradient, Cochran-Armitage) is descriptive context; a Mantel-Haenszel stratified test and logistic regression with region as a covariate assess the gradient robustly (§4.7); it is not used as a Bonferroni-survivor claim. Test 2b is a sensitivity variant of Test 2. Test 5 (founding classifier) is post-hoc (Appendix A.2). World religion sub-group analyses (§4.2) use Fisher exact tests and are not Bonferroni-corrected. Tests A, C, D, and E (§4.5) assess the 3° periodicity structure: Test A (full-corpus Rayleigh) is a boundary-condition test for invariant sinusoidal periodicity; Tests C and D are conditional anchor-specific tests; Test E is the anchor-discovery correction. Test B is tautological by construction and excluded from both tables.

3.6 Simulation Null Models

Three simulation approaches (100,000 trials each) and two dedicated geographic-concentration nulls are described with their results in §5.5.

3.7 Normalized Phasor Consensus (NPC) Test

The Normalized Phasor Consensus (NPC) test is a conditional anchor-concordance test at the candidate 3° period. It does not evaluate how the candidate frame was found; it asks whether related monument datasets, including externally curated corpora, yield similar anchor phases once that period is specified. The period $T = 3^\circ$ is fixed from the candidate-frame analysis, so NPC is a follow-up test at a stated candidate period rather than a blind search over periods. For each corpus c and candidate spacing T , site longitudes are mapped to phases on the T -periodic circle and summarized by the mean phasor

$$Z_c(T) = \frac{1}{N_c} \sum_{i=1}^{N_c} \exp(2\pi i \lambda_{ci}/T).$$

The corpus contributes only its direction, $\tilde{Z}_c(T) = Z_c(T)/|Z_c(T)|$, so NPC compares recovered anchors rather than giving larger or more concentrated corpora more

weight. For C corpora,

$$K(T) = \left| \frac{1}{C} \sum_{c=1}^C \tilde{Z}_c(T) \right|$$

ranges from 0 to 1, with larger values indicating tighter agreement among recovered anchor phases.

The permutation null preserves each corpus’s internal phasor magnitude but destroys cross-corpus phase agreement by independently rotating each corpus’s unit phasor by a uniform random angle. The resulting permutation distribution asks whether the observed directions are more concordant than expected if corpora retained their own internal structure but had unrelated anchors.

NPC and the full-corpus Rayleigh test address different alternatives. Rayleigh asks whether one corpus has a global sinusoidal periodicity in raw longitude at 3° . NPC asks whether several datasets yield similar anchor phases at a specified period after their phasors are normalized. A positive NPC result and a null Rayleigh result can therefore coexist; neither test substitutes for the other.

The monument NPC analysis uses the UNESCO Cultural/Mixed corpus ($N = 1011$), the context-validated UNESCO domed-monument subset ($N = 83$), and the Wikidata Q180987 stupa corpus ($N = 229$). These are three monument datasets, two externally assembled and one nested within the UNESCO corpus. The Wikidata corpus also informed the tier-threshold dispersion scale (§3.3); this should be noted for tier-based cross-corpus arguments, although the NPC statistic itself uses phasor directions rather than tier counts. OWTRAD is used as a contrast corpus at the same candidate period, with $N = 1674$ unique route nodes; degree-weighted OWTRAD endpoint positions are reported separately for within-frame enrichment.

3.8 Computational Reproducibility

All statistics are generated by an automated pipeline; every script writes to a shared results store and LaTeX macros are regenerated without manual editing. Scripts, audit files, and reproduction instructions are catalogued in Supplementary S1; see also §A.2).

4 Results

4.1 Boundary Conditions: What Is Not Being Claimed

Three checks delimit the scope of the present claim before the within-frame diagnostics are reported. They tell us what the result is *not*: it is not an invariant global 3° periodicity in raw longitude, and it is not a phase ranking that survives a full anchor-discovery correction. We treat these results as boundary conditions on the interpretation, not as a verdict on the selected-frame enrichment or on the NPC anchor-concordance result reported below.

(i) The full-corpus Rayleigh test returns null ($p_{\text{perm}} = 0.3716$ ns). The Rayleigh/periodogram null tells us what the result is not: it is not global sinusoidal periodicity in raw longitude at 3° . The conditional diagnostics ask a different question: whether sparse node-neighborhood overrepresentation is structured within the specified candidate frame. Lattice-node proximity and spectral periodicity are distinct mathematical objects, so a selected-frame count excess can coexist with a null Rayleigh statistic. The null Rayleigh result prevents an invariant-period interpretation.

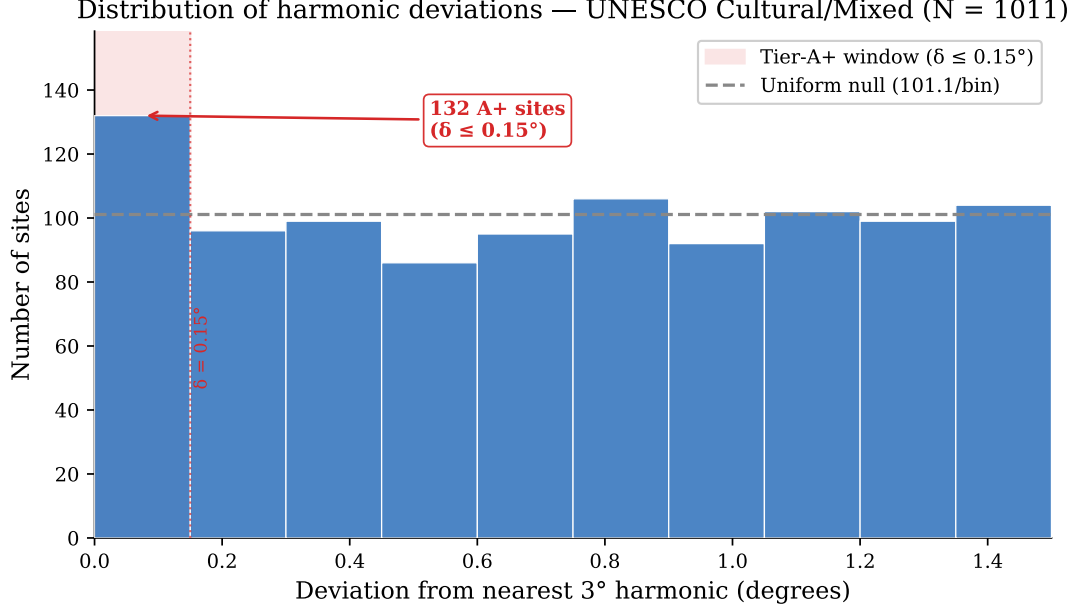


Figure 1: Distribution of harmonic deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.15^\circ$, shaded) above the uniform null (dashed line) is consistent with the corpus-level signal against which Tests 2 and 3 characterize morphological and spatial specificity.

(ii) A phase-envelope scan over all origins $\phi \in [0, 3)$ (300 phases at 0.01° resolution) records strong phase localization within the selected 3° circle: full-corpus A+ counts span 83–132 (median 100.0), and Gerizim’s phase ($2.2690 \bmod 3^\circ$) sits at the 100.0th percentile (132 A+), the 96.7th for domes (18/90; range 4–21). Within the selected representation, the chosen phase is unusually good; this is a within-frame diagnostic, not an invariant claim.

(iii) A degree-denominated Monte Carlo unit sweep identifies 3° as the dominant period (§6.2); adjacent non-harmonic units are null (§5.1). The unit sweep does not require an anchor and is reported here as the unit-side complement to the phase scan: together they indicate that 3° is the spacing at which the within-frame count excess concentrates and that the selected phase is the phase at which the concentration is sharpest, while the anchor-discovery correction across all anchors is marginal (Test E, $p = 0.0753$, [†]; §4.5).

(iv) NPC is reported as a separate anchor-concordance result rather than as a substitute for Rayleigh. Rayleigh tests global smooth periodicity within one corpus; NPC tests similarity of recovered anchors across datasets at the candidate period. The two results can coexist because they address different alternatives.

The remainder of the Results follows the evidentiary hierarchy. It first reports the primary conditional frame evidence in the full corpus, dome/stupa subset, and harmonic-band structure; it then reports NPC anchor concordance across related monument datasets and places robustness and exploratory diagnostics in separate sections. These sections are not presented as phase- or anchor-marginalized confirmation.

4.2 Primary Conditional Frame Evidence: Full-Corpus Sparse Enrichment (Test 1)

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 13.1% (Figure 1) (132/1011; Clopper-Pearson 95% CI [11.0%, 15.3%]), above the 10% geometric null, which assumes uniform longitude distribution even though the corpus does not satisfy that assumption; empirical-null results are reported alongside (binomial $p = 0.0011$). The observed enrichment is $1.31\times$, modest relative to the dome subpopulation ($2.00\times$) and the harmonic cluster asymmetry. For transparency, the non-primary Tests 1–3 sensitivity summary is reported in Table 4; the primary multiple-comparisons report is the full $k = 44$ audit in §5.7.

Region-stratified robustness. A region-specific empirical null (Fisher combination of per-region binomials) yields $p = 0.7978$ (ns); the pre-2000 cohort remains significant against the geometric null ($p = 0.0132$ *; §4.7). Both results are included in the FDR audit (Table 4).

World religion sites

This subgroup audit is exploratory/descriptive and is not part of the primary conditional frame evidence. Religion sub-populations are defined by keyword sets documented in the supplementary audit archive (Tenenbaum, 2026). A chi-squared test finds no significant A+ rate difference across the five main religion groups ($\chi^2 = 4.753$, $df = 4$, $p = 0.3136$, ns). Both Buddhism (20/71 A-tier, 28.2%; 19/71 C-band, 26.8%; $p = 0.0052$ **) and Hinduism (11/39 A-tier, 28.2%; 10/39 C-band, 25.6%; $p = 0.0324$ *) show a joint A-tier/C-band signal (multivariate hypergeometric); both persist in tradition-exclusive sites ($p = 0.0123$ *, $p = 0.0862$ †).

4.3 Dome/Stupa Subset and Geographic Conditioning (Test 2)

The keyword filter is described in §3.4 ($N = 90$).

Tier-A++ concentration (key result). 11 of 90 dome sites (12.2%, $3.06\times$ the 4% null) fall within Tier-A++ ($\delta \leq 0.06$, $\lesssim 6.66$ km). This concentration is significant across three test statistics: Fisher OR = 2.71, $p = 0.0077$ **; permutation $Z = 3.21$, $p = 0.004$ **; bootstrap $p = 0.006$ **. Dome A++ survives full Bonferroni correction at $k = 44$ (§5.7), but Null A locates the pattern in the dome/stupa sites' own longitude geography (§4.3).

Tier-A+ enrichment (non-primary sensitivity). 18/90 sites at Tier-A+ (20.0% [12.3, 29.8], $2.00\times$; Fisher OR = 1.77, $p = 0.0346$ *; binomial $p = 0.0032$ **). Tier-A: 29/90 (32.2%, $1.61\times$; Fisher $p = 0.0175$ *; binomial $p = 0.0043$ **). The χ^2 -uniform test ($p = 0.0732$, †) is consistent with concentration at harmonic nodes. Tier-B (middle zone): 45/90 sites; mean $\delta = 0.6586^\circ$.

Tier-A+ sites: Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out binomial check indicates no single site drives the result: removing any A+ site yields $N = 89$, A+ = 17, $p = 0.0067$. Leave-2-out (worst case) across all $\binom{18}{2}$ combinations: $p = 0.0133$. Leave-3-out (worst case): $p = 0.0253$; the evolution corpus (Test 2b) (§4.4) and the A++ concentration are more robust. The A++

leave-3-out worst case across all $\binom{11}{3} = 165$ triples yields $p = 0.0233$, remaining significant at $\alpha = 0.05$ (leave-2-out worst case $p = 0.0087$).

Context-validated robustness (Tests 2x). Sentence-level context validation reduces the dome corpus to $N = 83$ (7 false positives rejected; Appendix A.1). Test 2x Fisher exact (dome sub-pop vs. full corpus, $N = 1011$): A++ (11/83, $3.00\times$, $p = 0.0041$ **); A+ (16/83, $1.67\times$, $p = 0.0614$ †); A (26/83, $1.64\times$, $p = 0.0346$ *). This context validation improves keyword precision but does not remove the Null A result that the sub-population pattern is carried by dome/stupa longitude geography.

Keyword-stratified sensitivity (leave-stupa-out). Partitioning the dome population by triggering keyword shows that stupa/stupas ($n = 18$) has the strongest individual signal (A++ OR = 6.94, $p = 0.0057$ **), but dome/tholos/spherical alone ($n = 65$, no stupa keyword) is significant in its own stratum at A++ (OR = 2.93, $p = 0.0193$ *). Mound keywords (tumulus/barrow/kofun) show no enrichment ($n = 0$), consistent with the filter being morphologically discriminating. These keyword-stratified checks are subject to the same Null A geographic-conditioning result as the full dome result.

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 71% fall in 20° – 120° E versus 41% of the full corpus. Two targeted null models ask how much of the dome enrichment is carried by this subset geography (100,000 trials).

Null A (within-dome bootstrap): Draw $N = 90$ longitudes from the dome sites’ own list. Mean = 18.00 A+, $p = 0.5429$ (ns). This null locates the observed A+ count in the dome sites’ own longitude geography: the dome sites’ positions are concentrated near harmonic longitudes.

Null C (restricted geographic draw): Pool all full-corpus sites within $\pm W^\circ$ of any dome site and bootstrap the expected A+ count. Dome sites (18 A+, 20.0%) nominally exceed the broader footprint at all three windows tested: $\pm 2^\circ$ (mean = 11.14), $\pm 5^\circ$ (mean = 11.65), and $\pm 10^\circ$ (mean = 11.84). The panel annotations in Figure 2 report the corresponding nominal p-values ($p < 0.05$ at each window individually); none survive Bonferroni correction across the three windows ($k = 3$).

Reconciliation. Null A locates the signal in the dome/stupa sites’ own longitude distribution, while Null C asks whether the same longitude footprint alone produces comparable counts. It does not: the restricted background expects approximately 11–12 A+ hits rather than the observed 18. Because the Null C window results are nominally significant but do not survive correction across the three windows, this is evidence for a geographically embedded subset pattern rather than a stand-alone claim of morphology-specific placement.

In the global anchor sweep (§5.3), Gerizim scores 132 A+ (99.7th percentile); Jerusalem scores 128 (97.7th).

4.4 Exploratory Variant: Mound Evolution (Test 2b)

Extending Test 2 with earthen mound-stage keywords (see §3.4); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.

Stage-by-stage breakdown (Table 2):

The combined population ($N = 110$, A+ = 18.2%; Table 2) is consistent with Test 2. The mound stage does not reach significance (8.3%, OR = 0.60 $\times$, Fisher $p = 0.8430$, ns). The signal is carried by the stupa (A+ 33.3%, OR = 3.44 $\times$, $p = 0.0216$ *) and dome sub-stages (A+ 17.3%, OR = 1.44 $\times$, $p = 0.1663$ ns). Leave-one-out checks indicate stability. Leave-3-out (worst case, 1140 triples): $p = 0.0371$.

Null C — restricted geographic draw: dome sites vs. same-longitude corpus

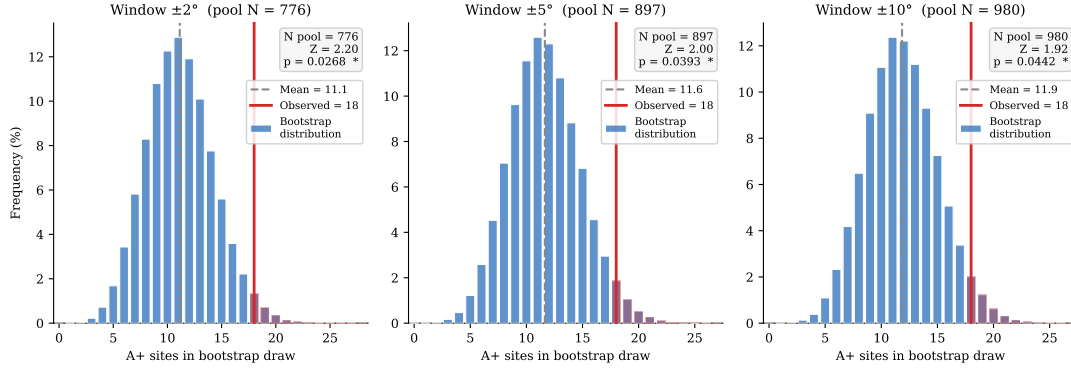


Figure 2: Null C bootstrap distributions for the dome sub-population ($N = 90$, observed $A+ = 18$) at three window widths. For each window $\pm W^\circ$, all full-corpus sites within $\pm W^\circ$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 90$ from that pool define the expected $A+$ distribution (blue bars). The observed count (red line) nominally exceeds the same-footprint background in all three windows; panel annotations report pool sizes, means, Z -scores, and p -values.

Table 2: Test 2b enrichment by evolutionary stage ($N = 110$). Null rates: $A++$ $P_0 = 4\%$, $A+$ $P_0 = 10\%$, A $P_0 = 20\%$. Fisher exact tests: stage vs. full corpus background.

Stage	N	Tier $A++$ (≤ 6.66 km)		Tier $A+$ (≤ 16.6 km)		Tier A (≤ 33 km)	
		$A++$	Rate OR (p)	$A+$	Rate OR (p)	A	Rate OR (p)
Mound ^a	24	2	8.3% $1.57\times$ (0.3880 ns)	2	8.3% $0.60\times$ (0.8430 ns)	6	25.0% $1.15\times$ (0.4653 ns)
Stupa ^b	18	4	22.2% $5.17\times$ (0.0144 *)	6	33.3% $3.44\times$ (0.0216 *)	7	38.9% $2.22\times$ (0.0874 $\dagger$)
Dome ^c	75	8	10.7% $2.21\times$ (0.0485 *)	13	17.3% $1.44\times$ (0.1663 ns)	23	30.7% $1.58\times$ (0.0576 $\dagger$)
Combined	110	13	11.8% $2.67\times$ (0.0049 **)	20	18.2% $1.57\times$ (0.0658 $\dagger$)	35	31.8% $1.71\times$ (0.0113 *)

^atumulus/barrow/kofun. ^bstupa/stupas. ^cdome/tholos/spherical. OR = odds ratio; p = one-sided Fisher exact vs. full UNESCO cultural corpus ($N = 1011$). Stage N values sum to more than the combined total because 4 sites match multiple stages.

Context-validated robustness (2bx). The context-validated 2bx sample remains robust: ($N = 104$, 6 rejected): $A++$ ($13/104$, $2.94\times$, $p = 0.0021$ **); $A+$ ($18/104$, $1.55\times$, $p = 0.0772$ $\dagger$); A ($32/104$, $1.61\times$, $p = 0.0239$ *).

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70° – 110° E). Null A (within-stupa bootstrap): $p = 0.5879$ (ns). Null C (restricted $\pm 5^\circ$ draw): $p = 0.0556$ ($\dagger$).

4.5 Primary Conditional Frame Evidence: Harmonic-Band / Cluster Asymmetry (Test 3)

Full corpus ($N = 1011$), no keyword filter (see §3.4). For each site, count other UNESCO sites within $\pm 0.30^\circ$ longitude; compare cluster sizes for $A+$ vs. non- $A+$ sites (Mann-Whitney U , permutation test). At the harmonic level, compare per-harmonic site counts for $A+$ -bearing vs. non- $A+$ harmonics.

$A+$ -bearing harmonics host $4.24\times$ more sites on average than non- $A+$ harmonics ($p < 0.0001$; Table 3). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.25 vs. 6.49 neighbors; $1.43\times$, $p = 0.0527$).

Table 3: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius $\pm 0.30^\circ$ lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within $\pm 0.30^\circ$ lon)	6.56	4.96	$1.32\times$	
Mann-Whitney U (one-tailed)				0.0025 **
Permutation (10,000 shuffles)			$Z = 3.96$	0.0000 ***
<i>Harmonic-level density (all Tier $\leq B$ sites per 3° harmonic):</i>				
Harmonics with ≥ 1 A+ site (36)	mean = 25.0 sites			
Harmonics with 0 A+ sites (19)	mean = 5.9 sites		$4.24\times$	< 0.0001 ***

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations) finds the conditional fraction of clustering-matched permutations producing the canonical-unit peak (0.1667) is identical to the unconditional rate (0.1786): the clustering and unit-sensitivity signals are statistically independent.

Rayleigh and anchor-specific checks. The A+-only Rayleigh statistic is omitted from the results because it is a mathematical consequence of the A+ classification criterion (§6.1), not an independent test. The following formal tests address distinct aspects of the periodicity:

Test A (full-corpus Rayleigh periodogram — boundary condition). The full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns) at $T = 3$ is null. We report this as a boundary condition, not a failure of the primary claim. The Rayleigh periodogram asks whether raw longitudes show a phase-marginalized sinusoidal periodicity across the corpus; it is structurally insensitive to sparse, anchor-specific clustering near a subset of fixed lattice nodes, because sites near nodes and anti-nodes are both treated as phase information about the same period. The present analysis therefore treats the Rayleigh null as evidence against invariant 3° periodicity, while leaving the selected-frame node-proximity count excess to be evaluated by the conditional tests below.

Test C (circular-shift anchor sensitivity). Displacing the anchor by a uniform random offset in $[0^\circ, 360^\circ)$ and recounting A+ sites yields a null mean of 101.05 vs. the observed 132 ($Z = 2.88$, $p_{\text{perm}} = 0.0061$, **). A positive anchor-specific test complementary to Test D.

Test D (targeted von Mises mixture, full corpus). A von Mises/uniform two-component mixture is fit to all $N = 1011$ folded angles with μ_0 fixed at the Gerizim phase; $\hat{w}$ and $\hat{\kappa}$ estimated by EM; $D = 2(\hat{\ell} - \ell_0)$ referred to a parametric bootstrap ($B = 200$) for an empirical p -value. On the full corpus: $D = 8.6103$, $\hat{w} = 0.0312$ (i.e. $\approx 3.1\%$ of all sites), $\hat{\kappa} = 36.70$ (angular SD $\approx 0.0788^\circ$), $p_{\text{boot}} = 0.0199$ (*). On the A+ subset alone: $D = 562.7225$, $\hat{w} = 1.0000$, $\hat{\kappa} = 31.89$ (angular SD $\approx 0.0845^\circ$), $p_{\text{boot}} = 0.0010$ (***). Tests C and D are positive anchor-specific checks within the selected 3° frame. They support the selected-frame characterization but do not convert it into a phase-free or whole-corpus periodicity claim.

Test E (anchor discovery correction). A global sweep of 36,000 trial anchors asks whether the Levantine corridor ranks unusually among all possible anchors ($p = 0.0753$, †). This is a conservative upper bound on the anchor-selection penalty; the strongest pattern rests on Tests 2 and 3: Test 2 is stable across the $x.25^\circ\text{E}$ co-grid

anchors, while Test 3 is keyword-independent but still characterizes the selected $3^\circ / \sim 35^\circ\text{E}$ representation (§6.1).

4.6 NPC Anchor Concordance Across Related Monument Corpora

At the candidate 3° period, the UNESCO Cultural/Mixed corpus ($N = 1011$), the context-validated UNESCO domed-monument subset ($N = 83$), and the Wikidata Q180987 stupa corpus ($N = 229$) yield similar recovered anchors: $\phi = +2.050^\circ$, $+2.027^\circ$, and $+1.868^\circ$, respectively. The three-dataset monument NPC is $K = 0.9857$ (perm- $p = 0.0119$, 100000 phase-randomization draws), and the consensus grid line falls 0.018° from 35°E (34.982°E).

NPC is not independent replication; it is cross-corpus anchor concordance within related Eurasian monument datasets. The statistic uses normalized phasor directions rather than tier counts, so it asks whether corpora recover similar anchors at the candidate period rather than whether they share the same A+/A++ counts. The UNESCO full corpus and the dome subset are nested, and Wikidata stupas are culturally and geographically related to the monument geography under study. The result therefore supports monument-corpus concordance under the NPC procedure, not geography-independent replication or a universal periodicity claim.

OWTRAD Silk Road route nodes provide a contrast at the same candidate period. Using 1674 unique nodes, OWTRAD’s own Rayleigh statistic at $T = 3^\circ$ is not significant ($R = 0.0308$, $p = 0.205$, ns), and its preferred anchor ($\phi = 2.75^\circ$) is 0.768° from the monument consensus. Including OWTRAD in the NPC calculation reduces consensus from $K = 0.9857$ to $K = 0.7713$ (perm- $p = 0.09182$, ns). The contrast is consistent with monument-corpus specificity and does not support a generic route-network signal, while still leaving shared Eurasian geography as a central feature of the interpretation.

4.7 Temporal Gradient (Test 4, Descriptive)

The Tier-A++ rate in the founding canon (1978–1984) is 8.0% (11/138; $1.38\times$ the 4% null, $p = 0.0232$ *), declining monotonically toward the null in later cohorts (2000–2025: 4.5%). Cochran-Armitage three-cohort trend test (A++, decreasing): $Z = -1.664$, $p = 0.0481$ (*). The five-cohort trend is marginal ($p = 0.0489$ *). By contrast, the A+ rate shows no significant decreasing trend ($p = 0.4184$ ns), indicating the gradient is specific to the highest-precision tier. The leading non-intentional explanation is a founding-canon selection effect: UNESCO’s earliest inscriptions prioritized famous monuments; famous monuments cluster in well-known cultural centers; and those centers are disproportionately located along the Eurasian corridor in which the selected 3° frame is being characterized. The Mantel-Haenszel stratified test is underpowered at the sub-group level; Logistic regression with region as a covariate is consistent with the A++ gradient ($Z = -1.760$, $p = 0.0784$ †, OR = 0.722). The declining rate may also partly reflect UNESCO’s expansion into regions where hemispherical traditions are less prevalent. The gradient is treated as descriptive context only (Figure 3).

4.8 FDR audit and Bonferroni correction

The primary multiple-comparisons headline is the full reported family: $k = 44$ tests in the FDR audit.

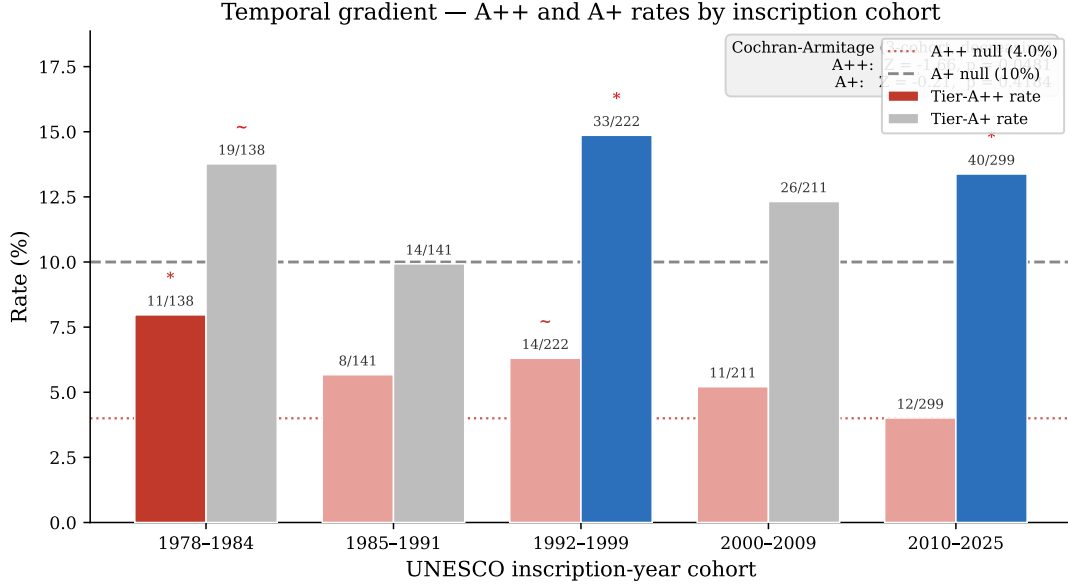


Figure 3: Tier-A++ and Tier-A+ rates by UNESCO inscription-year cohort. The founding canon (1978–1984) shows an elevated A++ rate (8.0%, $p = 0.0232$ *) that declines monotonically across later cohorts toward the 4% null (dotted line; Cochran-Armitage $Z = -1.664$, $p = 0.0481$ *). The A+ rate (right bars) shows no significant trend ($p = 0.4184$ ns).

Table 4: Non-primary Tests 1–3 sensitivity summary. Raw p-values are shown only to summarize the three headline conditional diagnostics. This grouping was not pre-specified as a multiplicity family; the primary multiple-comparisons report is the full $k = 44$ audit.

Test	N	Raw p	Role
Test 1 (full corpus)	1011	0.0011 ^a	Conditional count enrichment
Test 2 (dome A+)	90	0.0346 ^a (OR = 1.77)	Subset sensitivity
Test 3 (cluster asym.)	1011	0.0000 ^{b,c}	Harmonic-band structure
Test 4 (temporal)	1011	0.4184 ^b	Descriptive
Test 2b (mound evol.)	110	0.0658 ^a (OR = 1.57)	Sensitivity

^aFisher exact p vs. rest of corpus (one-sided, enrichment); ^bpermutation p ; ^c0.0000 denotes no exceedances at reported precision. Binomial baselines: Test 2 $p = 0.0032$ **, Test 2b $p = 0.0062$ **.

A comprehensive FDR audit (44 tests; [Benjamini & Hochberg 1995](#)) retains 22 at $q < 0.05$, of which 6 survive full Bonferroni at $k = 44$. The 6 full- k survivors are: dome/spherical A++ (binomial $p_{\text{adj}} = 0.0416$, with Null A locating the pattern in dome-site longitude geography); full-corpus A+ (binomial $p_{\text{adj}} = 0.0465$); cluster permutation ($p_{\text{adj}} = < 0.0001$); harmonic Mann-Whitney ($p_{\text{adj}} = 0.0012$); sequential-peak permutation ($p_{\text{adj}} = 0.0279$; §4.7); and evolution combined A++ (binomial $p_{\text{adj}} = 0.0208$).

Non-primary Tests 1–3 sensitivity summary

Table 4 reports raw p-values for Tests 1–3 as a sensitivity summary. This grouping is not treated as a separate multiplicity family; the full $k = 44$ audit is the primary multiple-comparisons report.

5 Robustness, Sensitivity, and Exploratory Diagnostics

5.1 Unit Sensitivity Sweep

Varying harmonic spacing from 1.5° to 6° (physical threshold fixed at 16.6 km), the 3° spacing produces the strongest joint significance across both the dome and full-corpus populations. The 1.5° and 6° spacings also reach significance, but both are mathematical consequences of the 3° structure: every 3° harmonic is also a 1.5° harmonic, and 6° harmonics are a strict subset, so neither constitutes an independent signal. Adjacent non-harmonic spacings (1.8° , 2.1° , 2.7° , 3.3° , 3.6°) are non-significant in either population, the key discriminating observation. The 2.4° spacing reaches nominal dome significance ($p = 0.0506$) but not full-corpus ($p = 0.2470$, ns), reflecting unsystematic local overlap with the 3° grid rather than a separate secondary spacing. Specifically, 7/16 (43.8%) of 2.4° dome hits align to the 3° grid, indicating the significance is due to overlap.

A degree-denominated Monte Carlo sweep (§6.2) is consistent with this pattern. Candidate intervals at 4° and 8° reach nominal dome significance, but inspection shows that 6 of 7 and all 4 of their hits respectively fall on 12° and 24° nodes shared with the 3° grid, indicating they are consequences of the selected 3° spacing rather than independent angular units.

5.2 Sensitivity Slope at the Canonical Unit

A fine sweep within $\pm 1\%$ of 3° shows a $\pm 0.3\%$ shift collapses joint significance in at least one population. A geographic bootstrap of the full cultural corpus finds no permutation in 10000 trials that reproduces the canonical-unit sharp peak at the A^+ precision tier ($p = 0.0000$, ***), consistent with 3° as the candidate spacing within the selected-frame audit. The dome-filter specificity test yields marginal results across all tiers (A^{++} : $p = 0.1728$, ns; A^+ : $p = 0.0999$, $\sim$; A : $p = 0.0427$, *), consistent with the test being underpowered at $n = 83$ sites. Restricting to the 22 morphologically unambiguous spherical forms (stupa, tholos) recovers significance at the A tier ($p = 0.0239$, *). Tests C and D (§4.5) are consistent with 3° structure.

A log-scale threshold sweep (73 values, 27/28 significant within ± 1 log-decade of A^+) yields dome-population data-optimal bands $A^{++} \approx 5.0$ km, $A^+ \approx 9.2$ km, $A \approx 17.0$ km (geometric ratio $r^* = 1.87$); the analytically derived tier boundaries (0.06° , 0.15° , 0.3°) are the nearest $\hat{\sigma}$ -multiple boundaries to these optima.

5.3 Global Anchor Sweep

A global sweep (0 – 360° E, 0.01° resolution, 36,000 trial anchors) assesses whether the $\sim 35^\circ$ E corridor is unusual globally. The sweep uses the extended corpus described in §3.2 ($N_{\text{ext}} = 1012$), each anchor excluding its own entry, so the focal anchors test against $N = 1011$.

Both anchors place the corridor in the global top 2.3% (36,000 trial anchors, 0 – 360° E; Gerizim: 132 A^+ , 99.7th percentile; Jerusalem: 128 A^+ , 97.7th percentile, and the result is anchor-invariant within the 4.1 km corridor window. The comparison anchors tested here, including Mecca, Mt. Meru, and Mt. Kailash, do not yield comparable enrichment.

The $x.25^\circ\text{E}$ periodic structure

Every $x.25^\circ\text{E}$ longitude (120 anchors) produces identically the maximum A+ count, spaced 3° apart. The label denotes phase $\approx 2.25^\circ \bmod 3^\circ$; Gerizim’s phase (2.269° , gap $0.019^\circ \approx 1.8\text{ km}$) is within coordinate uncertainty.

Formal statistical tests for the $x.25^\circ$ periodicity

The full-corpus Rayleigh ($p_{\text{perm}} = 0.3716$, ns) does not show global periodicity; Test B is tautological. Tests C ($p_{\text{perm}} = 0.0061$, **) and D ($p_{\text{boot}} = 0.0199$, *) are positive anchor-specific tests; Test E ($p = 0.0753$, $\dagger$, marginal) is the anchor discovery correction. The unit sensitivity sweep (§5.1) and Monte Carlo unit sweep (§6.2) are consistent with 3° as the dominant period without assuming any particular unit.

A dome-phase uniformity check does not distinguish alignment from geographic concentration. Gerizim scores at the 96.5th percentile among dome sites; all 18 dome A+ sites lie within ± 0.15 of the $x.25^\circ$ optimal phase.

Corridor and landmark ranking. Of the 13 A+ sites unique to Gerizim and 9 unique to Jerusalem, 119 are shared; the enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as anchor, Gerizim ranks #8 of 1012 (top 2.8%).

5.4 Spatial Autocorrelation

A+ harmonics host $4.24\times$ as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC ($\pm 0.5^\circ$, $\rho = 0.187$): DEFF = 1.842, $N_{\text{eff}} = 549$, corrected $p = 0.011$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI [10.50%, 15.43%] excludes the 10% null. Within-group permutation using UNESCO-region $\times$ longitude bins reproduced the observed A+ count exactly in every trial, showing this test cannot discriminate alignment from geographic concentration; the Null A and Null C simulations (§4.3) address that question directly.

5.5 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.060$ $\dagger$; 1978-1984 canon ($N = 138$) $p = 0.433$ ns; pre-2000 ($N = 501$) $p = 0.490$ ns. (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.070$ $\dagger$. (3) KDE-smoothed null: expected 101.1 ± 9.6 , observed 132, $Z = 3.23$, $p = 0.001$ **. The KDE null ($p = 0.001$ **) is consistent with full-corpus Tier-A+ enrichment; permutation and bootstrap are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.004$ **, bootstrap $p = 0.006$ **), with Null A locating the A++ pattern in dome-site longitude geography. The geographic-concentration nulls (Null A, Null C; §4.3) locate the pattern in dome/stupa monument geography and provide marginal same-footprint evidence for residual subset structure.

5.6 Related External Check: Wikidata Q180987 Stupa Corpus

The pipeline was applied to $N = 229$ Wikidata-attested stupas (class Q180987) as a within-method, geographically related external check using an independently curated, but geographically related, corpus (Tenenbaum, 2026). The Wikidata stupa corpus is concentrated in the Eurasian corridor, so the analysis is a related-corpus check

within the target geography rather than a test of geography-independent replication. The Kedu Plain (Java) hosts the densest cluster at the 75° harmonic (75°E from Gerizim), anchored by Borobudur (c. 800 CE, 7.23 km, Tier A+) (Miksic, 1990); the World Peace Pagoda at Lumbini achieves A++ precision (~ 555 m) under the Gerizim-labeled phase.

Using the identical anchor and null, the corpus yields 70/229 Tier-A sites (30.6%, $1.53\times$ null; binomial $p < 0.001$, ***; permutation $Z = 1.59$, $p_{\text{perm}} = 0.065$, $\sim$). No robust Tier-A+ enrichment is found beyond nominal binomial significance (binomial $p = 0.021$, *), consistent with cluster-level rather than individual-site precision; this result is not treated as a portability confirmation at the A+ threshold. Harmonic nodes with at least one Tier-A site carry $2.56\times$ more stupa sites on average than empty nodes, a related harmonic-band density pattern under the same representation.

Geographic tier breakdown and bimodal structure. All regional p-values below are one-sided Fisher exact tests. A sub-regional breakdown reveals a bimodal pattern consistent with the Buddhist and Hindu bimodal enrichment in the UNESCO religion audit (§4.2). The Java/Sumatra node (105°–115°E, near the 75° harmonic at 110.3°E) concentrates 43 of 83 sites at Tier-A (51.8%, OR = $4.74\times$, $p < 0.001$ ***), rising to 42/54 (77.8%, OR = $18.38\times$, $p < 0.001$ ***) in the 107°–112°E window, where A+ is also elevated at 31.5% (OR = $4.57\times$, $p < 0.001$ ***). The physical heartland, India/Nepal (70°–90°E) and Myanmar/Cambodia (90°–105°E), is A-tier depleted (combined OR = $0.21\times$, $p < 0.001$ ***, one-sided depletion test) but strongly enriched in the inter-harmonic C-band: Myanmar/Cambodia reaches 35.8% (OR = $4.47\times$, $p < 0.001$ ***) and the combined heartland 29.2% (OR = $6.00\times$, $p < 0.001$ ***). This A/C anti-correlation mirrors the bimodal pattern in the religion audit (§4.2). Non-heartland Wikidata stupas (outside 70°–110°E, $N = 102$) show the same A-enriched, C-depleted profile: 50.0% reach Tier-A (OR = $5.68\times$, $p < 0.001$ ***) with no C-band enrichment (OR = $0.19\times$, $p = 1.000$ ns).

5.7 Multiple Comparisons

See §3.5 for the full protocol. Gerizim ranks at the 99.7th percentile; a $\pm 0.3\%$ unit shift collapses joint significance. FDR audit (44 tests) retains 22 at $q < 0.05$; 6 survive Bonferroni at $k = 44$.

6 Limitations

6.1 Conditional Frame and Scope

Conditional status. The analysis is conditional on a stated candidate frame: 3° spacing, the Gerizim-labeled $\sim 35^\circ\text{E}$ phase, the tier thresholds, and the dome/stupa keyword definitions. The paper does not claim pre-registered or parameter-free discovery of this frame, and conditional p-values should not be read as phase-free discovery statistics. The relevant inferential questions are instead: (1) whether the specified frame organizes the externally curated UNESCO corpus in a statistically nontrivial way; (2) whether the subset signal remains structured under geographically conditioned nulls; (3) whether related externally curated monument corpora show anchor concordance at the same candidate period; and (4) whether negative controls delimit the scope of the pattern.

Anchor-specificity as a property of the selected representation. The within-frame organization characterized in §4.2–§4.5 is by construction anchor-specific: it is a property of the chosen 3° / $\sim 35^\circ\text{E}$ representation, not a claim about a phase that

survives marginalization over all anchors. The global anchor sweep (§5.3) is consistent with the enrichment being a property of the $\sim 35^\circ\text{E}$ corridor: every $x.25^\circ\text{E}$ anchor produces the same A+ count, and both Gerizim and Jerusalem place the corridor in the global top 2.3% of 36,000 trial anchors. Gerizim is distinguished within that corridor by pre-existing documentary evidence—the *tabbur ha-aretz* cosmographic role (§2.2)—predating this analysis. We do not present this as a unique solution to an inverse problem; we present Gerizim as one historically motivated point in a corridor whose width is itself part of the result.

The 3° structure is recoverable by any unanchored 120-phase scan within the corridor. Dome enrichment (Test 2) is identical across all 120 $x.25^\circ\text{E}$ anchors. Cluster asymmetry (Test 3) requires no keyword filter, but it is still defined by the same A+ classification and therefore characterizes the selected representation rather than supplying an anchor-independent test. The anchor-discovery correction (Test E, $p = 0.0753$, [†]) is marginal, which we report as a boundary condition (§4.1) on phase-free claims. It does not negate the conditional question of whether the specified frame organizes Eurasian monument data.

6.2 Geographic and Cultural Selection Bias

The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor of Cultural/Mixed sites. The regional distribution is as follows: (Europe & North America: 50%; Arab States: 9%; Asia-Pacific: 23%; Latin Americas: 11%; Africa: 7%). Domed/stupa-related monuments are part of this geography, not a population from which geographic structure must be removed before the signal can be meaningful. Null A ($p = 0.5429$, ns) shows that the observed dome/stupa enrichment is reproducible from the dome/stupa sites' own longitude distribution; the signal is carried by the subset geography itself. Null C asks a different question: whether the broader UNESCO background occupying the same longitude footprint produces comparable A+ enrichment. Across the tested windows the same-footprint background expects about 11.14–11.84 A+ hits rather than the observed 18; the resulting p-values are nominal and should be interpreted as marginal evidence that the subset's longitude structure exceeds same-footprint background expectations.

The cross-corpus evidence is likewise geographically conditioned. UNESCO Cultural/Mixed and Wikidata Q180987 stupas are external curations, but they are historically and geographically related. The UNESCO dome subset is nested within the UNESCO corpus. NPC therefore supports monument-corpus anchor concordance within Eurasian monument geography rather than independent replication outside that geography.

A Monte Carlo unit sweep (10000 permutations, 18 candidate spacing values from 0.5° to 15°) identifies which angular interval produces clustering beyond geographic concentration without assuming any particular unit. The dominant peak is at 3° ($p < 0.001$), with harmonics at 6° and 12° also significant. Collapsing harmonics, the dome subset yields one independent dominant spacing ($P = 0.2248$ under the geographic null), consistent with the selected-frame dome count excess concentrating at 3° rather than at adjacent candidate spacings.

6.3 Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows an A+ rate of 9.7%, below the 10% null (one-sided depletion: $p = 0.5149$ ns). The sample is too small to treat this as strong depletion, but it does not support extension of the pattern outside the

Eurasian footprint.

6.4 Other Limitations

Geocoding precision. UNESCO XML coordinates are administrative centroids, and nomination boundaries, serial-site representation, and corpus construction can affect site longitudes. Such noise is not expected to impose a common directional bias, but it limits the precision with which any node-neighborhood result should be interpreted.

Metrological evidence. A post-hoc numerical correspondence with Babylonian sub-units was noticed and discarded as non-load-bearing (Hunger & Steele, 2019; Robson, 2008; Powell, 1987); all thresholds are degree-denominated from cross-corpus angular dispersion.

These limitations define the claim’s scope rather than eliminating it. The paper reports a conditional, geographically embedded monument-corpus pattern under a specified representation; it does not claim metrology, intentional placement, universal periodicity, or a unique Gerizim origin.

7 Discussion

The empirical result is a sparse, structured node-neighborhood enrichment in Eurasian monument geography under a specified candidate 3° / $\sim 35^\circ\text{E}$ frame. The evidence comes from conditional selected-frame diagnostics and NPC monument-corpus anchor concordance, while the Rayleigh null, marginal anchor-discovery correction, geographic dependence, and OWTRAD contrast delimit what the result can mean.

7.1 Alternative Explanations

(1) Founding-canon and inscription selection. UNESCO’s founding canon prioritized famous monuments; famous monuments cluster in well-known cultural centers; and well-known cultural centers cluster along the Eurasian corridor. This selection pathway is a leading non-intentional explanation for the elevated early A++ rate reported in §4.7.

(2) Geographic concentration. Null A shows that the dome/stupa signal is embedded in the subset’s own longitude geography. This does not explain the signal away; it locates the signal in the geography of the monuments under study. Null C then compares that subset with the broader UNESCO background from the same longitude footprint and finds nominal residual overrepresentation (expected about 11.14–11.84 A+ hits versus observed 18). The Monte Carlo unit sweep (§6.2) is consistent with 3° without assuming any historical unit.

(3) Morphological recurrence. Hemispherical structures recur across cultures (Aveni, 2001, 2003), but recurrence by itself predicts no longitude specificity.

(4) Random coincidence. The KDE-smoothed null is consistent with dome enrichment ($p = 0.001^{**}$); permutation ($p = 0.060^\dagger$) and bootstrap ($p = 0.070^\dagger$) are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.004^{**}$; bootstrap $p = 0.006^{**}$), subject to the Null A geographic-conditioning result above. Tests C and D (§4.5) are positive anchor-specific checks within the selected frame, but they do not convert the conditional result into global periodicity.

7.2 Interpreting Modest Effect Sizes

The observed enrichment ($1.31\times$ full corpus, $2.00\times$ domes at A+, $3.06\times$ domes at A++, with the Null A result locating much of the sub-population pattern in dome-site geography) is internally consistent under several sensitivity checks but modest in magnitude. The effect is sparse: only a minority of sites fall near node neighborhoods. We interpret it as structured overrepresentation within Eurasian monument geography, not as strict observance of a 3° placement rule. Several explanations are consistent with this pattern:

(1) Founding-canon selection effects. UNESCO’s earliest inscriptions emphasized famous monuments. Famous monuments are disproportionately located in well-known cultural centers, and those centers are concentrated along the Eurasian corridor. This mechanism does not require intentional longitude placement and is given the same evidentiary weight here as logistical or diffusion explanations.

(2) Logistical convenience. Trade routes and way-stations naturally cluster at intervals determined by travel distance: one day’s journey (~ 30 km at 30° N latitude) corresponds to roughly 0.3° , so a 3° grid represents about ten such intervals and could capture higher-order staging posts without coordinated placement.

(3) Cultural diffusion with local variation. Buddhist architectural traditions spread along the Silk Roads with transmission hubs at specific longitudes (Lumbini, the Borobudur node at $\approx 110^\circ$ E), while local adaptation introduces scatter that flattens the effect size when averaged across the full corpus.

(4) Selective preservation beyond the founding canon. Later inscription and preservation processes may continue to favor culturally significant sites at trade junctions or political centers that happen to align with the corridor, even as the World Heritage List expands geographically.

These explanations remain speculative. The contribution of this paper is to document the conditional sparse-node signal, the NPC anchor-concordance result, and the geographically conditioned tests that delimit its interpretation; determining any historical mechanism would require independent lines of evidence beyond spatial statistics alone.

7.3 Cross-Corpus Anchor Concordance and Related-Corpus Checks

Rayleigh and NPC answer different questions. The full-corpus Rayleigh test ($R = 0.0224$, $p = 0.3716$, ns) asks whether UNESCO longitudes show global smooth sinusoidal periodicity at 3° . It does not. NPC asks whether several datasets yield similar recovered anchors at the candidate period after each corpus contributes a normalized phasor direction. The positive monument NPC result therefore changes the evidentiary balance, but it does not override the Rayleigh null or convert the selected-frame signal into an invariant-period claim.

Monument datasets and geographic confounding. As reported in §4.6, UNESCO Cultural/Mixed, the nested UNESCO domed-monument subset, and Wikidata Q180987 stupas yield similar NPC anchors at $T = 3^\circ$ ($K = 0.9857$, perm- $p = 0.0119$), with a consensus grid line 0.018° from 35° E. This is a real empirical observation within the tested monument datasets. UNESCO and Wikidata are independent in the curatorial sense relevant here: they are external databases and were not manually assembled to fit this candidate frame. They are not geography-independent, and the UNESCO dome subset is nested inside the UNESCO corpus. That limits causal or universal interpretations, but it does not undercut the stated

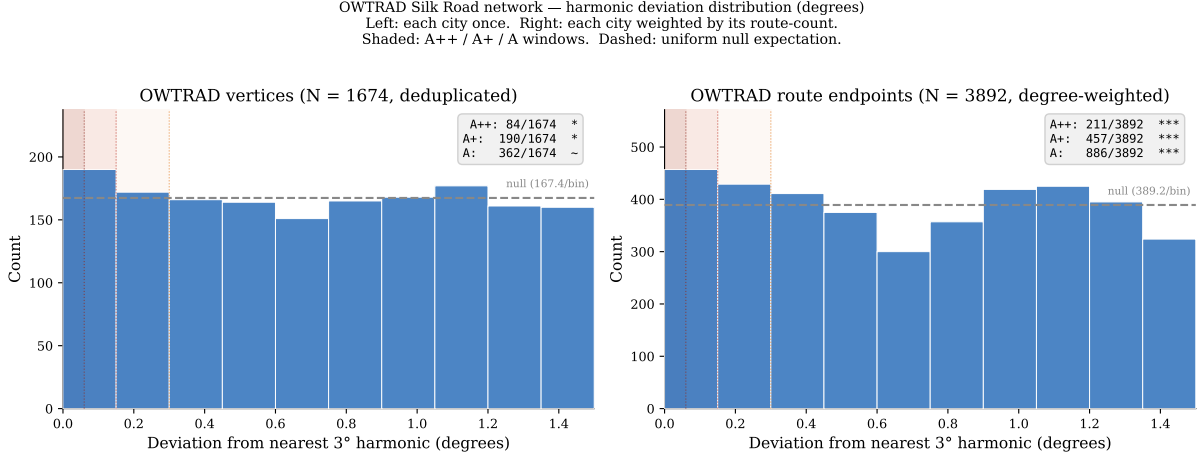


Figure 4: OWTRAD Silk Road endpoint tier distributions: 1674 deduplicated endpoints (left) and 3892 degree-weighted positions (right). Both show significant A++ enrichment (deduplicated: $z = 2.13$, $p = 0.0224$ *; degree-weighted: $z = 4.53$, $p = < 0.0001$ ***).

target: Eurasian monument geography.

OWTRAD as contrast corpus. OWTRAD Silk Road route nodes ($N = 1674$ unique nodes) do not share the monument NPC anchor: their preferred anchor is 0.768° from the monument consensus, and adding OWTRAD reduces NPC consensus to $K = 0.7713$ (perm- $p = 0.09182$, ns). This is consistent with monument-corpus specificity and does not support a generic route-network signal. It does not prove that shared Eurasian geography is irrelevant, since OWTRAD also lies largely within the same broad continental corridor and differs from the monument datasets in function, construction history, and sampling.

The within-frame OWTRAD enrichment results below answer a narrower question: whether OWTRAD nodes fall near the selected UNESCO grid lines when that grid is imposed. They are therefore related-corpus within-frame checks inside a shared Eurasian footprint, not evidence that OWTRAD recovers the monument anchor. The OWTRAD attested-route network (Ciolek, 2004) (1946 route segments, 1674 unique endpoints). Testing all 3892 endpoint positions across the full edge list (each city weighted by the number of segments it connects), 211 fall at A++ harmonics, 457 at A+, and 886 at A ($1.36\times$ null rate, $z = 4.53$, $p = < 0.0001$ *** for A++; $p = 0.0002$ *** for A+; $p = < 0.0001$ *** for A; one-tailed binomial). Testing each unique endpoint once gives 84/1674 at A++, 190 at A+, and 362 at A ($1.25\times$, $z = 2.13$, $p = 0.0224$ * for A++; $p = 0.0377$ * for A+; $p = 0.0524$ $\dagger$ for A; one-tailed binomial). A phase-shift test (random offset of the 3° grid, 100000 iterations) gives $p = 0.0202$ *, consistent with enrichment under the selected 3° / $\sim 35^\circ\text{E}$ phase when that frame is imposed. A cluster asymmetry test mirrors Test 3 of the primary analysis: harmonics containing at least one A++ endpoint (30 harmonics) carry $2.70\times$ more total route-connectivity than harmonics without one (11 harmonics; $p = 0.0007$ ***; Mann-Whitney one-tailed). The Silk Road corridor is illustrated in Figure 5; harmonic tier distributions for both OWTRAD populations are shown in Figure 4.

7.4 The Mechanism Gap

A literal reading of the within-frame organization—that builders across Eurasia placed monuments with sub-degree longitudinal precision relative to a common reference—

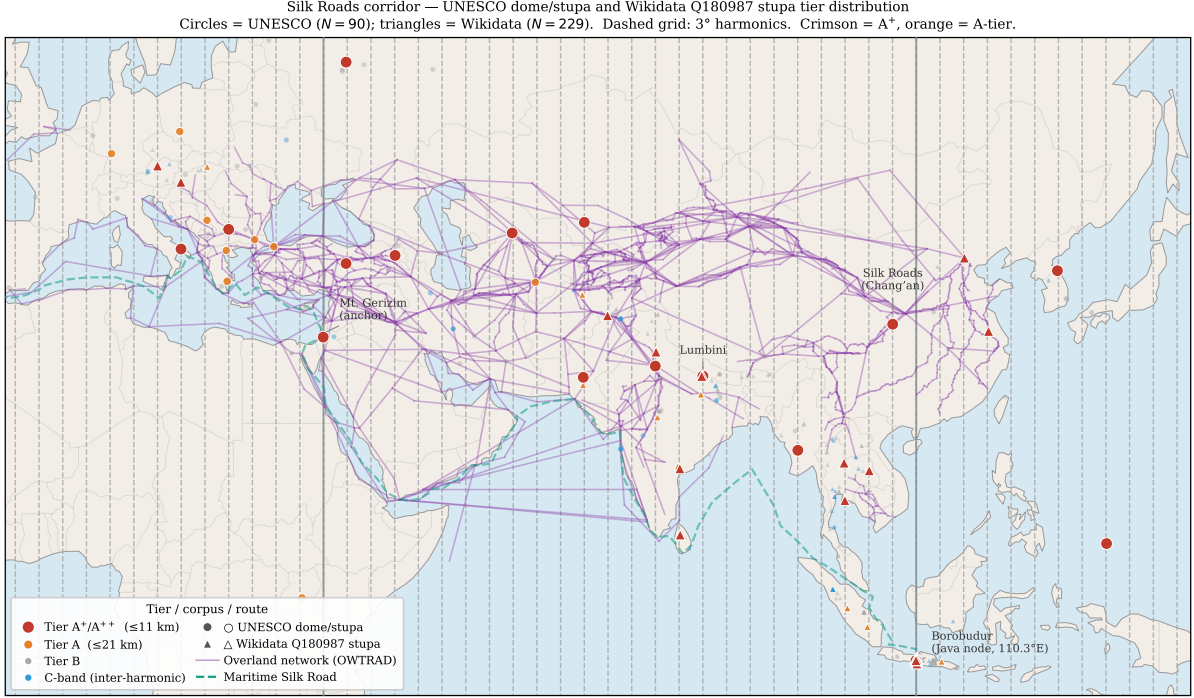


Figure 5: Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor. Overland route edges from the OWTRAD attested-route database (Ciolek, 2004). Maritime route is a schematic coastal polyline after Casson (1989), Sen (2014), and Tomber (2008). An interactive version is provided as supplementary material (supplementary/interactive_map/silkroad_interactive.html).

would require pre-modern continental longitude determination at a precision that is not a documented historical capability. Babylonian astronomical methods support angular measurement and eclipse timing (Hunger & Steele, 2019; Neugebauer, 1975; Robson, 2008) but do not extend to terrestrial longitude across continental scales, and no surveying instrument calibrated in ancient metrological sub-units relevant to such a procedure has been recovered. We take this gap seriously as a constraint on interpretation rather than as a peripheral caveat: any historical reading of the reported regularity must favor explanations that do not require coordinated placement.

This is why we frame the empirical contribution as conditional sparse-node enrichment within a specified representation rather than as evidence of metrology or coordinated design. The selected 3° / $\sim 35^\circ\text{E}$ / dome representation interacts with the actual geographic and cultural distribution of UNESCO sites—trade-route spacing at characteristic travel intervals, geographic channeling by river systems and passes, the concentration of inscribable cultural heritage along the Eurasian corridor, and inscription selection effects within UNESCO itself—to produce a set of within-frame characterizations. That the selected-frame count excess is sharper than chance under permutation and bootstrap nulls drawn from the same representation is the result we report. Whether the pattern ultimately reflects the geometry of Eurasian corridor geography under a particular representational choice, some monument-specific selection process, or another mechanism is not decided here, and we explicitly do not propose a historical or causal mechanism.

8 Conclusion

This paper identifies and audits a sparse, geographically embedded longitude-frame pattern in Eurasian monument corpora. Conditional on a specified post-discovery candidate $3^\circ / \sim 35^\circ\text{E}$ frame, the UNESCO Cultural/Mixed corpus shows modest Tier-A+ node-neighborhood enrichment, and monument subsets show stronger concentration. The effect is not whole-corpus alignment: most UNESCO sites do not fall near candidate harmonic nodes, and the full-corpus Rayleigh periodogram is null ($p = 0.3716$).

The dome/stupa findings should be read as structure within Eurasian monument geography, not as a pattern that must be independent of that geography. Null A ($p = 0.5429$) shows that the dome/stupa signal is reproducible from the subset's own longitude distribution. Null C asks whether the broader UNESCO background from the same longitude footprint produces comparable A+ enrichment; it expects about 11.14–11.84 A+ hits across the tested windows rather than the observed 18, giving marginal evidence for subset structure beyond same-footprint background expectations.

The NPC result adds monument-corpus anchor concordance at the candidate period, including concordance with an externally curated, geographically related stupa corpus. UNESCO Cultural/Mixed, the nested UNESCO domed-monument subset, and Wikidata Q180987 stupas yield a strong NPC statistic ($K = 0.9857$, $\text{perm-}p = 0.0119$), with a consensus grid line 0.018° from 35°E . UNESCO and Wikidata are independent curations in the relevant sense: they were not manually assembled to fit the candidate frame. They are not geography-independent, but geography-independence is not the claim being tested. The target, if real, is a structure within Eurasian monument geography itself. OWTRAD route nodes provide a useful negative control because they do not share the monument NPC anchor; including OWTRAD reduces consensus to $K = 0.7713$ ($\text{perm-}p = 0.09182$, ns), and OWTRAD's own 3° Rayleigh statistic is not significant ($p = 0.205$, ns).

The result is therefore best read as sparse, structured node-neighborhood enrichment within a specified $3^\circ / \sim 35^\circ\text{E}$ monument-corpus frame. It is supported by conditional diagnostics and monument-corpus anchor concordance, while being bounded by the null global periodogram, marginal anchor-discovery correction, geographic dependence, and OWTRAD non-portability. It is not evidence of universal periodicity, historical intentionality, whole-corpus alignment, metrology, or a unique Gerizim origin. Further work should retain the same standards used here: clear specification of the candidate frame, transparent conditional tests, externally curated corpora where available, negative controls, geographically conditioned null models, and explicit scope statements about Eurasian monument geography. We make all data, code, and methods freely available (§A.2; Zenodo DOI 10.5281/zenodo.19938283) so that those conditional claims can be inspected and reanalyzed.

A Keyword Audit Summary

Full site-by-site audit files are indexed in Supplementary S1 (supplementary/Supplementary_S1.md; Tenenbaum 2026). Regenerate all with `bash tools/update_all_audits.sh`.

A.1 Dome/Spherical Monument Filter (Tests 2, 2b, 2x, 2bx)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site’s name, XML description, and extended OUV text. All 90 matching sites are included in Test 2 without context gating.

For Test 2x, ambiguous keywords (*dome*, *domed*, *domes*, *spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural context. Unambiguous keywords (*stupa/stupas*, *tholos*) are accepted without validation. The seven rejected sites are: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuta, and Diquís Stone Spheres.

Test 2b adds earthen mound-stage keywords (*tumulus*, *tumuli*, *barrow*, *barrows*, *kofun* unambiguous; *mound* context-validated) to trace the hemispherical tradition from burial mound through stupa to architectural dome. Test 2bx applies the same architectural context gates to the extended population. Statistics for all four variants are reported in §4.3 and §4.4.

A.2 Founding-Origin Classifier (Test 5, Post-Hoc)

Results. The classifier finds 31.8% of A+ sites carry a founding keyword versus 28.7% in the full corpus (Fisher OR = 1.19, $p = 0.225$ ns). The A++ sub-population (56 sites) shows a stronger signal: 42.9% carry a founding keyword (OR = 1.94, $p = 0.014$ *), consistent with the most precisely aligned sites being disproportionately historically foundational monuments.

The classifier assigns sites to five founding-related categories; full keyword definitions and the site-by-site listing are in the supplementary archive (Tenenbaum, 2026).

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Data Availability Statement

All data and code are openly available; a permanent snapshot (v1.6.0) is deposited at Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19938283>); UNESCO XML: UNESCO World Heritage Centre 2024. Scripts, audit files, and maps are indexed in Supplementary S1 (supplementary/Supplementary_S1.md), with each file’s role, macros produced, and paper section noted. Full reproduction: `reproduce_all_macros.sh`, `update_all_audits.sh`, `generate_figures.py`.

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Conflict of Interest

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Software and AI Disclosure

All statistical analyses were performed in Python 3 using NumPy, SciPy, and pandas. Large-language-model tools were used during preparation of this manuscript: Claude (Anthropic) assisted with prose drafting and editing; GitHub Copilot (Microsoft) assisted with code autocompletion. These tools were used solely as writing and coding aids. All scientific hypotheses, analytical choices, keyword-list decisions, interpretation of results, and conclusions are solely the responsibility of the author. No AI tool was used to generate, fabricate, or substitute for primary data or statistical analysis.